Esquema IFC 4.3 (IFC4X3_ADD2)

Como o esquema é organizado, como ler EXPRESS, herança, atributos, relacionamentos, estrutura espacial de infraestrutura, tipos e regras, com dados extraídos do próprio esquema

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Gerado a partir do esquema IFC4X3_ADD2 embutido no IfcOpenShell 0.8.5 e da documentação-fonte oficial (repositório buildingSMART IFC4.3.x-development). Contagens, árvores e tabelas de atributos foram extraídas por código, não digitadas.

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1. Identidade do esquema

Item	Valor
Identificador no arquivo	FILE_SCHEMA(('IFC4X3_ADD2'));
Nome completo	IFC 4.3.2.0 (IFC4.3 Addendum 2)
Status normativo	ISO 16739-1:2024 (aprovada em janeiro de 2024)
Base	É o IFC4 (ISO 16739-1:2018) ampliado. Quase tudo do IFC4 continua valendo.
Escopo	Edificações + infraestrutura: rodovias, ferrovias, pontes, portos e hidrovias, terraplenagem e geotecnia.
Requisitos de troca	MVDs Reference View e Alignment-based Reference View; cada vez mais IDS (Information Delivery Specification)
Linguagem do esquema	EXPRESS (ISO 10303-11)
Formatos de arquivo	STEP (.ifc), ifcXML, .ifcZIP; ifcJSON em evolução
Tamanho	876 entidades, 132 tipos definidos, 243 enumerações, 61 SELECTs

> **Versões intermediárias:** IFC4X1 (alinhamento) e IFC4X2 (pontes) foram etapas de desenvolvimento e não devem ser usadas em produção. Arquivos antigos com FILE_SCHEMA('IFC4X3') (sem _ADD2) correspondem a versões anteriores do 4.3; o identificador atual é IFC4X3_ADD2.

2. O que é “o esquema”

O esquema IFC 4.3 é um arquivo escrito na linguagem **EXPRESS** (ISO 10303-11). Ele não contém nenhum projeto: é o dicionário que diz quais classes existem, quais atributos cada uma tem, em que ordem, de que tipo, quais são obrigatórios e quais regras precisam ser respeitadas. Um arquivo .ifc (STEP, ISO 10303-21) é só uma lista de instâncias desse dicionário. Por isso a primeira linha útil de qualquer .ifc diz qual dicionário usar: `FILE_SCHEMA(('IFC4X3_ADD2'))`.

Um esquema EXPRESS tem quatro tipos de declaração. A contagem abaixo foi feita diretamente no esquema embutido no IfcOpenShell 0.8.5:

Declaração	Qtd.	O que é	Exemplo real
ENTITY	876	Classe com atributos e herança. 133 são abstratas (não podem ser instanciadas) e 743 são concretas.	IfcWall, IfcProject, IfcRelAggregates

Declaração	Qtd.	O que é	Exemplo real
TYPE (tipo definido)	132	Apelido com significado para um tipo básico (STRING, REAL, INTEGER, BOOLEAN...).	IfcLabel = STRING; IfcLengthMeasure = REAL
TYPE ... ENUMERATION	243	Lista fechada de valores. No arquivo aparecem entre pontos.	IfcWallTypeEnum: .STANDARD., .USERDEFINED.
TYPE ... SELECT	61	"Um destes tipos": permite que um atributo aceite classes/tipos diferentes.	IfcValue, IfcAxis2Placement

Total: **1312 declarações**. Para comparação, dentro de IfcProduct (tudo que tem posição no espaço) há 214 entidades e dentro de IfcRelationship há 51.

3. Como ler uma declaração EXPRESS

Saber ler EXPRESS é o que permite abrir a documentação oficial e entender exatamente o que o arquivo aceita. As palavras-chave que importam:

Palavra-chave	Significado prático
`ENTITY ... END_ENTITY;`	Declara uma classe.
`ABSTRACT SUPERTYPE OF (ONEOF (A, B))`	A classe é abstrata e cada instância será exatamente de um dos subtipos listados (ONEOF).
`SUBTYPE OF (X)`	Herda todos os atributos de X, que vêm ANTES dos seus no arquivo.
`Nome : Tipo;`	Atributo obrigatório. No arquivo não pode ser $.
`Nome : OPTIONAL Tipo;`	Atributo opcional. Vazio = $.
`SET [1:?] OF X / LIST [2:3] OF X`	Coleção com cardinalidade mínima:máxima (? = sem limite). SET não tem ordem nem repetição; LIST tem ordem.
`INVERSE Nome : SET OF Rel FOR Atr;`	Atributo inverso: não é gravado no arquivo; é calculado a partir de quem aponta para esta instância.
`DERIVE`	Atributo calculado. Se redeclara um atributo herdado, no arquivo ele vira *.
`UNIQUE UR1 : GlobalId;`	Valor único entre todas as instâncias (ex.: GlobalId).
`WHERE WR1 : expressão;`	Regra de domínio que a instância precisa satisfazer. O leitor STEP não verifica; um validador sim.
`FUNCTION / RULE`	Funções auxiliares usadas nas regras e regras globais (entre várias entidades).

Declaração de IfcRoot no IFC4X3_ADD2 (reconstituída a partir do esquema carregado no IfcOpenShell; atributos, tipos e ordem são exatamente os do esquema):

```
ENTITY IfcRoot
 ABSTRACT SUPERTYPE OF (ONEOF
    (IfcObjectDefinition, IfcPropertyDefinition, IfcRelationship));
    GlobalId : IfcGloballyUniqueId;
    OwnerHistory : OPTIONAL IfcOwnerHistory;
    Name : OPTIONAL IfcLabel;
    Description : OPTIONAL IfcText;
END_ENTITY;
```

Leitura: IfcRoot é abstrata, tem três ramos (definições de objetos, definições de propriedades e relacionamentos) e define os quatro primeiros atributos de **toda** entidade com identidade. Repare que `OwnerHistory` é OPTIONAL: no 4.3 ele pode ser omitido ($), o que no 2x3 era proibido.

E a declaração de IfcWall:

```
ENTITY IfcWall
 SUPERTYPE OF (ONEOF
    (IfcWallStandardCase))
 SUBTYPE OF (IfcBuiltElement);
    PredefinedType : OPTIONAL IfcWallTypeEnum;
END_ENTITY;
```

4. Arquitetura em camadas e módulos

A norma organiza as declarações em quatro camadas. A regra de ouro: uma camada só pode referenciar a própria camada ou as de baixo. Isso mantém o núcleo estável e permite acrescentar domínios sem quebrar o resto. Cada camada é dividida em **módulos** (na documentação oficial, cada módulo é um capítulo).

Camada	Declarações	Papel
Domínio	344	Especializações de uma disciplina (HVAC, elétrica, estrutura, rodovia, ferrovia, portos).
Interoperabilidade (Shared)	162	Conceitos usados por várias disciplinas (paredes, lajes, elementos de serviço, elementos de infraestrutura).
Núcleo (Core)	190	Kernel e extensões: o esqueleto abstrato. Daqui para cima tudo herda de IfcRoot e tem GlobalId.
Recursos	616	Estruturas de apoio sem identidade própria (geometria, unidades, materiais, datas). Não existem sozinhas: são usadas por algo das camadas de cima.

Camada Domínio

Módulo	Decl.	Conteúdo
`IfcArchitectureDomain`	10	Propriedades pré-definidas de portas e janelas (lining/panel) e estilos/tipos arquitetônicos.
`IfcBuildingControlsDomain`	18	Automação predial: sensores, atuadores, controladores, alarmes.
`IfcConstructionMgmtDomain`	20	Recursos de construção: mão de obra, equipamentos, materiais de obra, subcontratos.
`IfcElectricalDomain`	75	Elétrica: quadros, cabos, luminárias, tomadas, motores, geradores, transformadores.
`IfcHvacDomain`	99	Climatização e ventilação: dutos, tubos, bombas, chillers, válvulas, conexões e seus tipos.
`IfcPlumbingFireProtectionDomain`	15	Hidrossanitário e combate a incêndio: louças, sprinklers, interceptores.
`IfcPortsAndWaterwaysDomain`	16	Novo no 4.3: portos e hidrovias (IfcMarineFacility, IfcMarinePart, IfcMooringDevice, IfcNavigationElement…).
`IfcRailDomain`	10	Novo no 4.3: ferrovia (IfcRailway, IfcRailwayPart, IfcRail, IfcTrackElement…).
`IfcRoadDomain`	7	Novo no 4.3: rodovia (IfcRoad, IfcRoadPart e enumerações próprias).
`IfcStructuralAnalysisDomain`	39	Modelo analítico estrutural: membros e conexões analíticas, apoios, cargas, grupos de carregamento, resultados.
`IfcStructuralElementsDomain`	35	Elementos estruturais físicos: fundações (sapatas, estacas), armaduras, cabos de protensão.

Camada Interoperabilidade (Shared)

Módulo	Decl.	Conteúdo
`IfcSharedBldgElements`	69	Elementos construtivos de edificação: paredes, lajes, vigas, pilares, portas, janelas, escadas, telhados, coberturas, e seus tipos.
`IfcSharedBldgServiceElements`	30	Base comum de instalações (MEP): elementos de distribuição, portas de conexão (ports), sistemas, propriedades de fluxo.

Módulo	Decl.	Conteúdo
IfcSharedComponentElements	15	Componentes discretos e fixadores: IfcFastener, IfcMechanicalFastener, IfcDiscreteAccessory.
IfcSharedFacilitiesElements	11	Mobiliário, ativos e inventário (IfcFurniture, IfcAsset, IfcInventory, IfcOccupant).
IfcSharedInfrastructureElements	27	Novo no 4.3: elementos comuns à infraestrutura (IfcCourse, IfcPavement, IfcKerb, IfcEarthworksCut/Fill, IfcReinforcedSoil, IfcSign, IfcSignal, IfcBearing, IfcDeepFoundation…).
IfcSharedMgmtElements	10	Gestão: custos (IfcCostItem, IfcCostSchedule), ordens de serviço, solicitações.

Camada Núcleo (Core)

Módulo	Decl.	Conteúdo
IfcControlExtension	4	Especializa os objetos de controle (IfcControl): aprovações, restrições, relações de controle.
IfcKernel	59	O “esqueleto” abstrato: IfcRoot, IfcObjectDefinition, IfcObject, IfcTypeObject, IfcProject, IfcRelationship e as relações genéricas (IfcRelAggregates, IfcRelDefinesBy*, IfcRelAssigns*, IfcRelAssociates*).
IfcProcessExtension	19	Processos e tarefas (IfcTask, IfcProcedure, IfcEvent), sequências e cronogramas.
IfcProductExtension	108	Produtos físicos e espaciais genéricos: IfcElement, IfcSpatialStructureElement (Site, Building, Storey, Space), IfcOpeningElement, IfcGrid, IfcAnnotation e as relações de contenção, aberturas e conexões.

Camada Recursos

Módulo	Decl.	Conteúdo
IfcActorResource	11	Pessoas, organizações e endereços (IfcPerson, IfcOrganization, IfcPostalAddress).
IfcApprovalResource	3	Aprovações (IfcApproval) e seus relacionamentos.
IfcConstraintResource	10	Restrições e métricas (IfcConstraint, IfcObjective, IfcMetric).
IfcCostResource	5	Valores de custo e valores aplicados (IfcCostValue, IfcAppliedValue).
IfcDateTimeResource	27	Datas e tempo. No 2x3 são entidades (IfcCalendarDate, IfcLocalTime, IfcDateAndTime); no IFC4+ são strings ISO 8601 e entidades de agenda/séries temporais.
IfcExternalReferenceResource	20	Referências externas: classificações, documentos, bibliotecas (IfcClassification, IfcDocumentInformation, IfcLibraryReference).
IfcGeometricConstraintResource	24	Posicionamento de produtos: IfcLocalPlacement, IfcGridPlacement (e IfcLinearPlacement no 4.3), geometria de conexão.
IfcGeometricModelResource	52	Modelos de sólidos e superfícies: extrusões, revoluções, B-rep, CSG/booleanas, malhas.
IfcGeometryResource	90	Geometria básica: pontos, direções, sistemas de eixos (IfcAxis2Placement*), curvas e superfícies.
IfcMaterialResource	22	Materiais: IfcMaterial, conjuntos de camadas (IfcMaterialLayerSet) e, no IFC4+, perfis e constituintes.
IfcMeasureResource	123	Tipos de medida (IfcLengthMeasure, IfcAreaMeasure…), unidades (IfcSIUnit, IfcConversionBasedUnit) e os SELECTs de valor.
IfcPresentationAppearanceResource	68	Estilos de apresentação: superfícies, curvas, hachuras, texto, texturas.
IfcPresentationDefinitionResource	8	Itens de anotação/apresentação: textos, símbolos, ocorrências de anotação.
IfcPresentationOrganizationResource	14	Camadas de apresentação (layers) e fontes de luz.
IfcProfileResource	32	Perfis 2D usados em extrusões (retângulo, I, T, circular, arbitrário…).

Módulo	Decl.	Conteúdo
IfcPropertyResource	16	Tipos de propriedade: IfcPropertySingleValue, IfcPropertyEnumeratedValue, IfcPropertyListValue, IfcComplexProperty…
IfcQuantityResource	10	Quantidades: IfcQuantityLength, IfcQuantityArea, IfcQuantityVolume, IfcQuantityCount, IfcQuantityWeight…
IfcRepresentationResource	26	Representações: IfcShapeRepresentation, contextos (IfcGeometricRepresentationContext), mapeamentos; no IFC4+ também georreferenciamento (IfcMapConversion, IfcProjectedCRS).
IfcStructuralLoadResource	26	Cargas estruturais (forças, momentos, deslocamentos, temperatura).
IfcTopologyResource	21	Topologia: vértices, arestas, loops, faces, shells (base para B-rep).
IfcUtilityResource	8	Utilitários: IfcOwnerHistory, IfcApplication, IfcTable.

5. Árvore de herança

Abaixo de IfcRoot o esquema se divide em três ramos. Saber em qual ramo uma classe está diz o que ela é: uma "coisa" (IfcObjectDefinition), uma ligação (IfcRelationship) ou informação anexada (IfcPropertyDefinition). Árvore gerada diretamente do esquema; "[+N subtipos]" indica quantas classes existem abaixo daquele nó:

```
IfcRoot (ABSTRACT)
├─ IfcObjectDefinition (ABSTRACT)
│  ├─ IfcContext (ABSTRACT)
│  │  ├─ IfcProject
│  │  └─ IfcProjectLibrary
│  ├─ IfcObject (ABSTRACT)
│  │  ├─ IfcActor  [+1 subtipos]
│  │  ├─ IfcControl (ABSTRACT)  [+10 subtipos]
│  │  ├─ IfcGroup  [+12 subtipos]
│  │  ├─ IfcProcess (ABSTRACT)  [+3 subtipos]
│  │  ├─ IfcProduct (ABSTRACT)
│  │  │  ├─ IfcAnnotation
│  │  │  ├─ IfcElement (ABSTRACT)  [+158 subtipos]
│  │  │  ├─ IfcLinearElement  [+4 subtipos]
│  │  │  ├─ IfcPort (ABSTRACT)  [+1 subtipos]
│  │  │  ├─ IfcPositioningElement (ABSTRACT)  [+4 subtipos]
│  │  │  ├─ IfcSpatialElement (ABSTRACT)  [+19 subtipos]
│  │  │  ├─ IfcStructuralActivity (ABSTRACT)  [+10 subtipos]
│  │  │  └─ IfcStructuralItem (ABSTRACT)  [+9 subtipos]
│  │  └─ IfcResource (ABSTRACT)  [+7 subtipos]
│  └─ IfcTypeObject  [+157 subtipos]
├─ IfcPropertyDefinition (ABSTRACT)
│  ├─ IfcPropertySetDefinition (ABSTRACT)  [+10 subtipos]
│  └─ IfcPropertyTemplateDefinition (ABSTRACT)  [+4 subtipos]
└─ IfcRelationship (ABSTRACT)  [+50 subtipos]
```

O ramo que mais interessa para manipular modelos é IfcElement → IfcBuiltElement, onde ficam os elementos físicos construídos:

```
IfcElement (ABSTRACT)
├─ IfcBuiltElement
│  ├─ IfcBeam
│  ├─ IfcBearing
│  ├─ IfcBuildingElementProxy
│  ├─ IfcChimney
│  ├─ IfcColumn
│  ├─ IfcCourse
│  ├─ IfcCovering
│  ├─ IfcCurtainWall
│  ├─ IfcDeepFoundation  [+2 subtipos]
```

```
│  ├─ IfcDoor
│  ├─ IfcEarthworksElement  [+2 subtipos]
│  ├─ IfcFooting
│  ├─ IfcKerb
│  ├─ IfcMember
│  ├─ IfcMooringDevice
│  ├─ IfcNavigationElement
│  ├─ IfcPavement
│  ├─ IfcPlate
│  ├─ IfcRail
│  ├─ IfcRailing
│  ├─ IfcRamp
│  ├─ IfcRampFlight
│  ├─ IfcRoof
│  ├─ IfcShadingDevice
│  ├─ IfcSlab
│  ├─ IfcStair
│  ├─ IfcStairFlight
│  ├─ IfcTrackElement
│  ├─ IfcWall  [+1 subtipos]
│  └─ IfcWindow
├─ IfcCivilElement
├─ IfcDistributionElement  [+82 subtipos]
├─ IfcElementAssembly
├─ IfcElementComponent (ABSTRACT)  [+14 subtipos]
├─ IfcFeatureElement (ABSTRACT)  [+7 subtipos]
├─ IfcFurnishingElement  [+2 subtipos]
├─ IfcGeographicElement
├─ IfcGeotechnicalElement (ABSTRACT)  [+5 subtipos]
├─ IfcTransportationDevice (ABSTRACT)  [+2 subtipos]
└─ IfcVirtualElement
```

Cadeia completa de herança de IfcWall:

```
IfcWall <- IfcBuiltElement <- IfcElement <- IfcProduct <- IfcObject <- IfcObjectDefinition <- IfcRoot
```

Na prática: `model.by_type("IfcBuiltElement")` devolve todas as instâncias de todas as subclasses (paredes, lajes, vigas...). `by_type("IfcWall")` também inclui IfcWallStandardCase. Use `include_subtypes=False` para pegar só a classe exata.

6. Atributos: a ordem é tudo

No arquivo STEP os atributos não têm nome, só posição. A posição é dada pela cadeia de herança: primeiro os atributos de IfcRoot, depois os de IfcObject, IfcProduct e assim por diante até a classe final. Tabela completa de IfcWall no IFC 4.3:

#	Atributo	Tipo	Preenchimento	Definido em
1	GlobalId	IfcGloballyUniqueId	obrigatório	IfcRoot
2	OwnerHistory	IfcOwnerHistory	opcional ($)	IfcRoot
3	Name	IfcLabel	opcional ($)	IfcRoot
4	Description	IfcText	opcional ($)	IfcRoot
5	ObjectType	IfcLabel	opcional ($)	IfcObject
6	ObjectPlacement	IfcObjectPlacement	opcional ($)	IfcProduct
7	Representation	IfcProductRepresentation	opcional ($)	IfcProduct
8	Tag	IfcIdentifier	opcional ($)	IfcElement
9	PredefinedType	IfcWallTypeEnum	opcional ($)	IfcWall

A mesma parede escrita no arquivo de exemplo deste documento (seção 12):

```
#28=IFCWALL('1VISKZXADCF9jqhay1nExU',$,'PAR-01',$,$,#27,#25,$,.SOLIDWALL.);
```

Lendo da esquerda para a direita, cada valor corresponde a uma linha da tabela: GlobalId, OwnerHistory, Name, Description ($), ObjectType ($), ObjectPlacement (referência para o IfcLocalPlacement), Representation (referência para a geometria), Tag ($) e PredefinedType (.SOLIDWALL.).

7. Entidades-chave e seus atributos

Tabelas com os atributos próprios (sem os herdados de IfcRoot, que são sempre GlobalId, OwnerHistory, Name e Description) das entidades que aparecem em praticamente todo arquivo.

IfcProject

Raiz do modelo: contextos de representação e unidades. Cadeia: IfcProject <- IfcContext <- IfcObjectDefinition <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	ObjectType	IfcLabel	opcional ($)
6	LongName	IfcLabel	opcional ($)
7	Phase	IfcLabel	opcional ($)
8	RepresentationContexts	SET [1:?] OF IfcRepresentationContext	opcional ($)
9	UnitsInContext	IfcUnitAssignment	opcional ($)

IfcSite

Terreno. Continua existindo, mas o georreferenciamento real fica em IfcMapConversion. Cadeia: IfcSite <- IfcSpatialStructureElement <- IfcSpatialElement <- IfcProduct ...

#	Atributo	Tipo	Preenchimento
5	ObjectType	IfcLabel	opcional ($)
6	ObjectPlacement	IfcObjectPlacement	opcional ($)
7	Representation	IfcProductRepresentation	opcional ($)
8	LongName	IfcLabel	opcional ($)
9	CompositionType	IfcElementCompositionEnum	opcional ($)
10	RefLatitude	IfcCompoundPlaneAngleMeasure	opcional ($)
11	RefLongitude	IfcCompoundPlaneAngleMeasure	opcional ($)
12	RefElevation	IfcLengthMeasure	opcional ($)
13	LandTitleNumber	IfcLabel	opcional ($)
14	SiteAddress	IfcPostalAddress	opcional ($)

IfcBuildingStorey

Pavimento. Cadeia: IfcBuildingStorey <- IfcSpatialStructureElement <- IfcSpatialElement <- IfcProduct ...

#	Atributo	Tipo	Preenchimento
5	ObjectType	IfcLabel	opcional ($)
6	ObjectPlacement	IfcObjectPlacement	opcional ($)
7	Representation	IfcProductRepresentation	opcional ($)
8	LongName	IfcLabel	opcional ($)
9	CompositionType	IfcElementCompositionEnum	opcional ($)
10	Elevation	IfcLengthMeasure	opcional ($)

IfcRelAggregates

Decomposição todo → partes (Projeto → Terreno → Edifício → Pavimento). Cadeia: IfcRelAggregates <- IfcRelDecomposes <- IfcRelationship <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	RelatingObject	IfcObjectDefinition	obrigatório
6	RelatedObjects	SET [1:?] OF IfcObjectDefinition	obrigatório

IfcRelContainedInSpatialStructure

Em qual nível espacial o elemento "mora". Cadeia: IfcRelContainedInSpatialStructure <- IfcRelConnects <- IfcRelationship <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	RelatedElements	SET [1:?] OF IfcProduct	obrigatório
6	RelatingStructure	IfcSpatialElement	obrigatório

IfcRelDefinesByType

Liga ocorrências ao seu tipo ("família"). Cadeia: IfcRelDefinesByType <- IfcRelDefines <- IfcRelationship <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	RelatedObjects	SET [1:?] OF IfcObject	obrigatório
6	RelatingType	IfcTypeObject	obrigatório

IfcRelDefinesByProperties

Liga um Pset/Qto aos objetos. Cadeia: IfcRelDefinesByProperties <- IfcRelDefines <- IfcRelationship <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	RelatedObjects	SET [1:?] OF IfcObjectDefinition	obrigatório
6	RelatingPropertyDefinition	IfcPropertySetDefinitionSelect	obrigatório

IfcPropertySet

Conjunto de propriedades (Pset_...). Cadeia: IfcPropertySet <- IfcPropertySetDefinition <- IfcPropertyDefinition <- IfcRoot

#	Atributo	Tipo	Preenchimento
5	HasProperties	SET [1:?] OF IfcProperty	obrigatório

IfcPropertySingleValue

Uma propriedade com um valor tipado. Cadeia: IfcPropertySingleValue <- IfcSimpleProperty <- IfcProperty <- IfcPropertyAbstraction

#	Atributo	Tipo	Preenchimento
1	Name	IfcIdentifier	obrigatório
2	Specification	IfcText	opcional ($)
3	NominalValue	IfcValue	opcional ($)
4	Unit	IfcUnit	opcional ($)

IfcLocalPlacement

Posição relativa ao objeto-pai. Cadeia: IfcLocalPlacement <- IfcObjectPlacement

#	Atributo	Tipo	Preenchimento
1	PlacementRelTo	IfcObjectPlacement	opcional ($)
2	RelativePlacement	IfcAxis2Placement	obrigatório

IfcShapeRepresentation

Uma representação geométrica (Body, Axis...). Cadeia: IfcShapeRepresentation <- IfcShapeModel <- IfcRepresentation

#	Atributo	Tipo	Preenchimento
1	ContextOfItems	IfcRepresentationContext	obrigatório
2	RepresentationIdentifier	IfcLabel	opcional ($)
3	RepresentationType	IfcLabel	opcional ($)
4	Items	SET [1:?] OF IfcRepresentationItem	obrigatório

IfcExtrudedAreaSolid

Sólido extrudado: perfil + direção + profundidade. Cadeia: IfcExtrudedAreaSolid <- IfcSweptAreaSolid <- IfcSolidModel <- IfcGeometricRepresentationItem ...

#	Atributo	Tipo	Preenchimento
1	SweptArea	IfcProfileDef	obrigatório
2	Position	IfcAxis2Placement3D	opcional ($)
3	ExtrudedDirection	IfcDirection	obrigatório
4	Depth	IfcPositiveLengthMeasure	obrigatório

IfcWallType

Tipo de parede. Cadeia: IfcWallType <- IfcBuiltElementType <- IfcElementType <- IfcTypeProduct ...

#	Atributo	Tipo	Preenchimento
5	ApplicableOccurrence	IfcIdentifier	opcional ($)
6	HasPropertySets	SET [1:?] OF IfcPropertySetDefinition	opcional ($)
7	RepresentationMaps	LIST [1:?] OF IfcRepresentationMap	opcional ($)
8	Tag	IfcLabel	opcional ($)
9	ElementType	IfcLabel	opcional ($)
10	PredefinedType	IfcWallTypeEnum	obrigatório

IfcAlignment

Alinhamento (eixo de obra linear). Cadeia: IfcAlignment <- IfcLinearPositioningElement <- IfcPositioningElement <- IfcProduct ...

#	Atributo	Tipo	Preenchimento
5	ObjectType	IfcLabel	opcional ($)
6	ObjectPlacement	IfcObjectPlacement	opcional ($)
7	Representation	IfcProductRepresentation	opcional ($)
8	PredefinedType	IfcAlignmentTypeEnum	opcional ($)

IfcLinearPlacement

Posicionamento ao longo de um eixo (estaca/afastamento). Cadeia: IfcLinearPlacement <- IfcObjectPlacement

#	Atributo	Tipo	Preenchimento
1	PlacementRelTo	IfcObjectPlacement	opcional ($)
2	RelativePlacement	IfcAxis2PlacementLinear	obrigatório
3	CartesianPosition	IfcAxis2Placement3D	opcional ($)

IfcPointByDistanceExpression

Ponto definido por distância ao longo de uma curva. Cadeia: IfcPointByDistanceExpression <- IfcPoint <- IfcGeometricRepresentationItem <- IfcRepresentationItem

#	Atributo	Tipo	Preenchimento
1	DistanceAlong	IfcCurveMeasureSelect	obrigatório
2	OffsetLateral	IfcLengthMeasure	opcional ($)
3	OffsetVertical	IfcLengthMeasure	opcional ($)
4	OffsetLongitudinal	IfcLengthMeasure	opcional ($)
5	BasisCurve	IfcCurve	obrigatório

IfcRoadPart

Parte de rodovia (pista, acostamento, interseção...). Cadeia: IfcRoadPart <- IfcFacilityPart <- IfcSpatialStructureElement <- IfcSpatialElement ...

#	Atributo	Tipo	Preenchimento
5	ObjectType	IfcLabel	opcional ($)
6	ObjectPlacement	IfcObjectPlacement	opcional ($)
7	Representation	IfcProductRepresentation	opcional ($)
8	LongName	IfcLabel	opcional ($)
9	CompositionType	IfcElementCompositionEnum	opcional ($)
10	UsageType	IfcFacilityUsageEnum	obrigatório
11	PredefinedType	IfcRoadPartTypeEnum	opcional ($)

IfcMapConversion

Georreferenciamento: local → sistema projetado. Cadeia: IfcMapConversion <- IfcCoordinateOperation

#	Atributo	Tipo	Preenchimento
1	SourceCRS	IfcCoordinateReferenceSystemSelect	obrigatório
2	TargetCRS	IfcCoordinateReferenceSystem	obrigatório
3	Eastings	IfcLengthMeasure	obrigatório
4	Northings	IfcLengthMeasure	obrigatório
5	OrthogonalHeight	IfcLengthMeasure	obrigatório
6	XAxisAbscissa	IfcReal	opcional ($)
7	XAxisOrdinate	IfcReal	opcional ($)
8	Scale	IfcReal	opcional ($)

8. Atributos inversos

Os inversos não aparecem no arquivo: são a "volta" das relações. É por eles que a API navega de um objeto para as relações que apontam para ele. Todos os inversos disponíveis em IfcWall:

Inverso	Relação	Atributo da relação	Card.	Definido em
HasAssignments	IfcRelAssigns	RelatedObjects	SET [0:?]	IfcObjectDefinition
Nests	IfcRelNests	RelatedObjects	SET [0:1]	IfcObjectDefinition
IsNestedBy	IfcRelNests	RelatingObject	SET [0:?]	IfcObjectDefinition
HasContext	IfcRelDeclares	RelatedDefinitions	SET [0:1]	IfcObjectDefinition

Inverso	Relação	Atributo da relação	Card.	Definido em
IsDecomposedBy	IfcRelAggregates	RelatingObject	SET [0:?]	IfcObjectDefinition
Decomposes	IfcRelAggregates	RelatedObjects	SET [0:1]	IfcObjectDefinition
HasAssociations	IfcRelAssociates	RelatedObjects	SET [0:?]	IfcObjectDefinition
IsDeclaredBy	IfcRelDefinesByObject	RelatedObjects	SET [0:1]	IfcObject
Declares	IfcRelDefinesByObject	RelatingObject	SET [0:?]	IfcObject
IsTypedBy	IfcRelDefinesByType	RelatedObjects	SET [0:1]	IfcObject
IsDefinedBy	IfcRelDefinesByProperties	RelatedObjects	SET [0:?]	IfcObject
ReferencedBy	IfcRelAssignsToProduct	RelatingProduct	SET [0:?]	IfcProduct
PositionedRelativeTo	IfcRelPositions	RelatedProducts	SET [0:?]	IfcProduct
ReferencedInStructures	IfcRelReferencedInSpatialStructure	RelatedElements	SET [0:?]	IfcProduct
FillsVoids	IfcRelFillsElement	RelatedBuildingElement	SET [0:1]	IfcElement
ConnectedTo	IfcRelConnectsElements	RelatingElement	SET [0:?]	IfcElement
IsInterferedByElements	IfcRelInterferesElements	RelatedElement	SET [0:?]	IfcElement
InterferesElements	IfcRelInterferesElements	RelatingElement	SET [0:?]	IfcElement
HasProjections	IfcRelProjectsElement	RelatingElement	SET [0:?]	IfcElement
HasOpenings	IfcRelVoidsElement	RelatingBuildingElement	SET [0:?]	IfcElement
IsConnectionRealization	IfcRelConnectsWithRealizingElements	RealizingElements	SET [0:?]	IfcElement
ProvidesBoundaries	IfcRelSpaceBoundary	RelatedBuildingElement	SET [0:?]	IfcElement
ConnectedFrom	IfcRelConnectsElements	RelatedElement	SET [0:?]	IfcElement
ContainedInStructure	IfcRelContainedInSpatialStructure	RelatedElements	SET [0:1]	IfcElement
HasCoverings	IfcRelCoversBldgElements	RelatingBuildingElement	SET [0:?]	IfcElement
HasSurfaceFeatures	IfcRelAdheresToElement	RelatingElement	SET [0:?]	IfcElement

Exemplo: `parede.ContainedInStructure[0].RelatingStructure` devolve o pavimento; `parede.IsDefinedBy` devolve as relações com Psets; o tipo chega por `parede.IsTypedBy` (inverso separado desde o IFC4).

9. Relacionamentos (IfcRelationship)

Toda ligação entre objetos é uma entidade própria. O IFC 4.3 tem 51 classes de relacionamento. Árvore completa:

```
IfcRelationship (ABSTRACT)
├─ IfcRelAssigns (ABSTRACT)
│  ├─ IfcRelAssignsToActor
│  ├─ IfcRelAssignsToControl
│  ├─ IfcRelAssignsToGroup
│  │  └─ IfcRelAssignsToGroupByFactor
│  ├─ IfcRelAssignsToProcess
│  ├─ IfcRelAssignsToProduct
│  └─ IfcRelAssignsToResource
├─ IfcRelAssociates (ABSTRACT)
│  ├─ IfcRelAssociatesApproval
│  ├─ IfcRelAssociatesClassification
│  ├─ IfcRelAssociatesConstraint
```

```
│  ├─ IfcRelAssociatesDocument
│  ├─ IfcRelAssociatesLibrary
│  ├─ IfcRelAssociatesMaterial
│  └─ IfcRelAssociatesProfileDef
├─ IfcRelConnects (ABSTRACT)
│  ├─ IfcRelConnectsElements
│  │  ├─ IfcRelConnectsPathElements
│  │  └─ IfcRelConnectsWithRealizingElements
│  ├─ IfcRelConnectsPortToElement
│  ├─ IfcRelConnectsPorts
│  ├─ IfcRelConnectsStructuralActivity
│  ├─ IfcRelConnectsStructuralMember
│  │  └─ IfcRelConnectsWithEccentricity
│  ├─ IfcRelContainedInSpatialStructure
│  ├─ IfcRelCoversBldgElements
│  ├─ IfcRelCoversSpaces
│  ├─ IfcRelFillsElement
│  ├─ IfcRelFlowControlElements
│  ├─ IfcRelInterferesElements
│  ├─ IfcRelPositions
│  ├─ IfcRelReferencedInSpatialStructure
│  ├─ IfcRelSequence
│  ├─ IfcRelServicesBuildings
│  └─ IfcRelSpaceBoundary
│     └─ IfcRelSpaceBoundary1stLevel
│        └─ IfcRelSpaceBoundary2ndLevel
├─ IfcRelDeclares
├─ IfcRelDecomposes (ABSTRACT)
│  ├─ IfcRelAdheresToElement
│  ├─ IfcRelAggregates
│  ├─ IfcRelNests
│  ├─ IfcRelProjectsElement
│  └─ IfcRelVoidsElement
└─ IfcRelDefines (ABSTRACT)
   ├─ IfcRelDefinesByObject
   ├─ IfcRelDefinesByProperties
   ├─ IfcRelDefinesByTemplate
   └─ IfcRelDefinesByType
```

Assinatura de cada relação concreta (atributos após os quatro de IfcRoot). O padrão é sempre o mesmo: um lado **Relating** (singular) e um lado **Related** (coleção):

Relação	Atributos próprios
`IfcRelAdheresToElement`	`RelatingElement: IfcElement, RelatedSurfaceFeatures: SET OF IfcSurfaceFeature`
`IfcRelAggregates`	`RelatingObject: IfcObjectDefinition, RelatedObjects: SET OF IfcObjectDefinition`
`IfcRelAssignsToActor`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingActor: IfcActor, ActingRole: IfcActorRole`
`IfcRelAssignsToControl`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingControl: IfcControl`
`IfcRelAssignsToGroup`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingGroup: IfcGroup`
`IfcRelAssignsToGroupByFactor`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingGroup: IfcGroup, Factor: IfcRatioMeasure`
`IfcRelAssignsToProcess`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingProcess: IfcProcessSelect, QuantityInProcess: IfcMeasureWithUnit`
`IfcRelAssignsToProduct`	`RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingProduct: IfcProductSelect`

Relação	Atributos próprios
IfcRelAssignsToResource	RelatedObjects: SET OF IfcObjectDefinition, RelatedObjectsType: IfcStrippedOptional, RelatingResource: IfcResourceSelect
IfcRelAssociatesApproval	RelatedObjects: SET OF IfcDefinitionSelect, RelatingApproval: IfcApproval
IfcRelAssociatesClassification	RelatedObjects: SET OF IfcDefinitionSelect, RelatingClassification: IfcClassificationSelect
IfcRelAssociatesConstraint	RelatedObjects: SET OF IfcDefinitionSelect, Intent: IfcLabel, RelatingConstraint: IfcConstraint
IfcRelAssociatesDocument	RelatedObjects: SET OF IfcDefinitionSelect, RelatingDocument: IfcDocumentSelect
IfcRelAssociatesLibrary	RelatedObjects: SET OF IfcDefinitionSelect, RelatingLibrary: IfcLibrarySelect
IfcRelAssociatesMaterial	RelatedObjects: SET OF IfcDefinitionSelect, RelatingMaterial: IfcMaterialSelect
IfcRelAssociatesProfileDef	RelatedObjects: SET OF IfcDefinitionSelect, RelatingProfileDef: IfcProfileDef
IfcRelConnectsElements	ConnectionGeometry: IfcConnectionGeometry, RelatingElement: IfcElement, RelatedElement: IfcElement
IfcRelConnectsPathElements	ConnectionGeometry: IfcConnectionGeometry, RelatingElement: IfcElement, RelatedElement: IfcElement, RelatingPriorities: LIST [0:?] OF IfcInteger, RelatedPriorities: LIST [0:?] OF IfcInteger, RelatedConnectionType: IfcConnectionTypeEnum, RelatingConnectionType: IfcConnectionTypeEnum
IfcRelConnectsPortToElement	RelatingPort: IfcPort, RelatedElement: IfcDistributionElement
IfcRelConnectsPorts	RelatingPort: IfcPort, RelatedPort: IfcPort, RealizingElement: IfcElement
IfcRelConnectsStructuralActivity	RelatingElement: IfcStructuralActivityAssignmentSelect, RelatedStructuralActivity: IfcStructuralActivity
IfcRelConnectsStructuralMember	RelatingStructuralMember: IfcStructuralMember, RelatedStructuralConnection: IfcStructuralConnection, AppliedCondition: IfcBoundaryCondition, AdditionalConditions: IfcStructuralConnectionCondition, SupportedLength: IfcLengthMeasure, ConditionCoordinateSystem: IfcAxis2Placement3D
IfcRelConnectsWithEccentricity	RelatingStructuralMember: IfcStructuralMember, RelatedStructuralConnection: IfcStructuralConnection, AppliedCondition: IfcBoundaryCondition, AdditionalConditions: IfcStructuralConnectionCondition, SupportedLength: IfcLengthMeasure, ConditionCoordinateSystem: IfcAxis2Placement3D, ConnectionConstraint: IfcConnectionGeometry
IfcRelConnectsWithRealizingElements	ConnectionGeometry: IfcConnectionGeometry, RelatingElement: IfcElement, RelatedElement: IfcElement, RealizingElements: SET OF IfcElement, ConnectionType: IfcLabel
IfcRelContainedInSpatialStructure	RelatedElements: SET OF IfcProduct, RelatingStructure: IfcSpatialElement
IfcRelCoversBldgElements	RelatingBuildingElement: IfcElement, RelatedCoverings: SET OF IfcCovering
IfcRelCoversSpaces	RelatingSpace: IfcSpace, RelatedCoverings: SET OF IfcCovering
IfcRelDeclares	RelatingContext: IfcContext, RelatedDefinitions: SET OF IfcDefinitionSelect
IfcRelDefinesByObject	RelatedObjects: SET OF IfcObject, RelatingObject: IfcObject
IfcRelDefinesByProperties	RelatedObjects: SET OF IfcObjectDefinition, RelatingPropertyDefinition: IfcPropertySetDefinitionSelect
IfcRelDefinesByTemplate	RelatedPropertySets: SET OF IfcPropertySetDefinition, RelatingTemplate: IfcPropertySetTemplate
IfcRelDefinesByType	RelatedObjects: SET OF IfcObject, RelatingType: IfcTypeObject
IfcRelFillsElement	RelatingOpeningElement: IfcOpeningElement, RelatedBuildingElement: IfcElement
IfcRelFlowControlElements	RelatedControlElements: SET OF IfcDistributionControlElement, RelatingFlowElement: IfcDistributionFlowElement
IfcRelInterferesElements	RelatingElement: IfcInterferenceSelect, RelatedElement: IfcInterferenceSelect, InterferenceGeometry: IfcConnectionGeometry, InterferenceType: IfcIdentifier, ImpliedOrder: IfcLogical, InterferenceSpace: IfcSpatialZone
IfcRelNests	RelatingObject: IfcObjectDefinition, RelatedObjects: LIST OF IfcObjectDefinition
IfcRelPositions	RelatingPositioningElement: IfcPositioningElement, RelatedProducts: SET OF IfcProduct

Relação	Atributos próprios
IfcRelProjectsElement	RelatingElement: IfcElement, RelatedFeatureElement: IfcFeatureElementAddition
IfcRelReferencedInSpatialStructure	RelatedElements: SET OF IfcSpatialReferenceSelect, RelatingStructure: IfcSpatialElement
IfcRelSequence	RelatingProcess: IfcProcess, RelatedProcess: IfcProcess, TimeLag: IfcLagTime, SequenceType: IfcSequenceEnum, UserDefinedSequenceType: IfcLabel
IfcRelServicesBuildings	RelatingSystem: IfcSystem, RelatedBuildings: SET OF IfcSpatialElement
IfcRelSpaceBoundary	RelatingSpace: IfcSpaceBoundarySelect, RelatedBuildingElement: IfcElement, ConnectionGeometry: IfcConnectionGeometry, PhysicalOrVirtualBoundary: IfcPhysicalOrVirtualEnum, InternalOrExternalBoundary: IfcInternalOrExternalEnum
IfcRelSpaceBoundary1stLevel	RelatingSpace: IfcSpaceBoundarySelect, RelatedBuildingElement: IfcElement, ConnectionGeometry: IfcConnectionGeometry, PhysicalOrVirtualBoundary: IfcPhysicalOrVirtualEnum, InternalOrExternalBoundary: IfcInternalOrExternalEnum, ParentBoundary: IfcRelSpaceBoundary1stLevel
IfcRelSpaceBoundary2ndLevel	RelatingSpace: IfcSpaceBoundarySelect, RelatedBuildingElement: IfcElement, ConnectionGeometry: IfcConnectionGeometry, PhysicalOrVirtualBoundary: IfcPhysicalOrVirtualEnum, InternalOrExternalBoundary: IfcInternalOrExternalEnum, ParentBoundary: IfcRelSpaceBoundary1stLevel, CorrespondingBoundary: IfcRelSpaceBoundary2ndLevel
IfcRelVoidsElement	RelatingBuildingElement: IfcElement, RelatedOpeningElement: IfcFeatureElementSubtraction

10. Estrutura espacial

```
IfcSpatialElement (ABSTRACT)
├─ IfcExternalSpatialStructureElement (ABSTRACT)
│  └─ IfcExternalSpatialElement
├─ IfcSpatialStructureElement (ABSTRACT)
│  ├─ IfcBuildingStorey
│  ├─ IfcFacility
│  │  ├─ IfcBridge
│  │  ├─ IfcBuilding
│  │  ├─ IfcMarineFacility
│  │  ├─ IfcRailway
│  │  └─ IfcRoad
│  ├─ IfcFacilityPart (ABSTRACT)
│  │  ├─ IfcBridgePart
│  │  ├─ IfcFacilityPartCommon
│  │  ├─ IfcMarinePart
│  │  ├─ IfcRailwayPart
│  │  └─ IfcRoadPart
│  ├─ IfcSite
│  └─ IfcSpace
└─ IfcSpatialZone
```

A grande mudança do 4.3 é a generalização da árvore espacial. **IfcFacility** (instalação) é a superclasse de IfcBuilding, IfcRoad, IfcRailway, IfcBridge e IfcMarineFacility. Cada instalação se divide em **IfcFacilityPart** (IfcRoadPart, IfcRailwayPart, IfcBridgePart, IfcMarinePart ou a genérica IfcFacilityPartCommon), cujo atributo obrigatório `UsageType` (IfcFacilityUsageEnum: LATERAL, LONGITUDINAL, VERTICAL, REGION…) diz como a parte divide a instalação. IfcBuildingStorey continua sendo a parte específica de edifícios.

IfcSpatialZone (desde o IFC4) permite zonas espaciais fora da hierarquia rígida, e **IfcExternalSpatialElement** representa o "exterior" (usado em limites de espaço). Alinhamentos (IfcAlignment) não são elementos espaciais: são IfcPositioningElement, agregados ao IfcProject e referenciados (IfcRelReferencedInSpatialStructure) nas instalações, nunca "contidos".

```
IfcProject
└─ IfcSite
   ├─ IfcBuilding ── IfcBuildingStorey ── IfcSpace
   ├─ IfcRoad ── IfcRoadPart
   ├─ IfcRailway ── IfcRailwayPart
   ├─ IfcBridge ── IfcBridgePart
   ├─ IfcMarineFacility ── IfcMarinePart
   └─ (qualquer IfcFacility) ── IfcFacilityPartCommon

Ligações: IfcRelAggregates (espacial) · IfcRelContainedInSpatialStructure (elementos)
IfcAlignment --IfcRelAggregates--> IfcProject
             --IfcRelReferencedInSpatialStructure--> IfcRoad / IfcRailway
```

10.1 Posicionamento: o ramo IfcPositioningElement

```
IfcPositioningElement (ABSTRACT)
├─ IfcGrid
├─ IfcLinearPositioningElement
│  └─ IfcAlignment
└─ IfcReferent
```

O alinhamento tem duas camadas: a **semântica** (IfcAlignment aninha, via IfcRelNests, IfcAlignmentHorizontal, IfcAlignmentVertical e IfcAlignmentCant, cada um com seus IfcAlignmentSegment e parâmetros de projeto) e a **geométrica** (IfcCompositeCurve, IfcGradientCurve ou IfcSegmentedReferenceCurve feitas de IfcCurveSegment). Elementos são posicionados ao longo do eixo com IfcLinearPlacement + IfcAxis2PlacementLinear + IfcPointByDistanceExpression, e ligados ao posicionador por IfcRelPositions.

11. Tipos de dados, enumerações e regras

11.1 Tipos definidos

Tipo	Tipo base	Observação
IfcGloballyUniqueId	STRING	22 caracteres (GUID de 128 bits comprimido).
IfcLabel	STRING	Texto curto (até 255).
IfcText	STRING	Texto longo.
IfcIdentifier	STRING	
IfcLengthMeasure	REAL	
IfcPositiveLengthMeasure	IfcLengthMeasure	Tipo derivado de outro tipo, com regra WHERE (> 0).
IfcRatioMeasure	REAL	
IfcBoolean	BOOLEAN	
IfcLogical	LOGICAL	.T., .F. ou .U. (desconhecido).
IfcTimeStamp	INTEGER	Segundos desde 1970-01-01 (Unix).
IfcDateTime	STRING	Ex.: '2026-09-28T10:00:00'.
IfcDate	STRING	

11.2 SELECTs: o caso de IfcValue

Valores de propriedades (NominalValue de IfcPropertySingleValue) são do tipo IfcValue, um SELECT de SELECTs. Como o atributo aceita vários tipos, o arquivo precisa dizer qual foi usado: por isso aparece `IFCLABEL('EI 60')` e não só `'EI 60'`.

Membro de IfcValue	Qtd.	Tipos aceitos (amostra)
IfcDerivedMeasureValue	71	IfcAbsorbedDoseMeasure, IfcAccelerationMeasure, IfcAngularVelocityMeasure, IfcAreaDensityMeasure, IfcCompoundPlaneAngleMeasure, IfcCurvatureMeasure, IfcDoseEquivalentMeasure, IfcDynamicViscosityMeasure, IfcElectricCapacitanceMeasure, IfcElectricChargeMeasure, IfcElectricConductanceMeasure, IfcElectricResistanceMeasure, IfcElectricVoltageMeasure, IfcEnergyMeasure …
IfcMeasureValue	23	IfcAmountOfSubstanceMeasure, IfcAreaMeasure, IfcComplexNumber, IfcContextDependentMeasure, IfcCountMeasure, IfcDescriptiveMeasure, IfcElectricCurrentMeasure, IfcLengthMeasure, IfcLuminousIntensityMeasure, IfcMassMeasure, IfcNonNegativeLengthMeasure, IfcNormalisedRatioMeasure, IfcNumericMeasure, IfcParameterValue …
IfcSimpleValue	15	IfcBinary, IfcBoolean, IfcDate, IfcDateTime, IfcDuration, IfcIdentifier, IfcInteger, IfcLabel, IfcLogical, IfcPositiveInteger, IfcReal, IfcText, IfcTime, IfcTimeStamp …

11.3 Enumerações e PredefinedType

No 4.3, PredefinedType existe em **153 classes de ocorrência** e 132 classes de tipo. A regra geral (WHERE CorrectPredefinedType): se PredefinedType = .USERDEFINED., o atributo ObjectType precisa ser preenchido com o texto do tipo. Se houver um tipo associado, o PredefinedType do tipo prevalece.

Valores de IfcWallTypeEnum no IFC 4.3: `.ELEMENTEDWALL.`, `.MOVABLE.`, `.PARAPET.`, `.PARTITIONING.`, `.PLUMBINGWALL.`, `.POLYGONAL.`, `.RETAININGWALL.`, `.SHEAR.`, `.SOLIDWALL.`, `.STANDARD.`, `.WAVEWALL.`, `.USERDEFINED.`, `.NOTDEFINED.`.

11.4 Regras (WHERE)

As regras WHERE são verificadas por validadores, não pelo leitor. Regras de algumas entidades-chave no 4.3 (nome da regra e o que exige, conforme a documentação oficial):

Entidade	Regra	Exigência
IfcWall	CorrectPredefinedType	PredefinedType vazio (ex.: porque há IfcWallType associado) ou, se for .USERDEFINED., ObjectType tem de estar preenchido.
IfcWall	CorrectTypeAssigned	Se houver tipo associado (inverso IsTypedBy), ele tem de ser um IfcWallType.
IfcWallStandardCase	HasMaterialLayerSetUsage	Exige um IfcMaterialLayerSetUsage associado (mesma exigência do WR1 no 2x3).
IfcRelAggregates	NoSelfReference	O RelatingObject não pode aparecer dentro de RelatedObjects (nada se agrega a si mesmo).
IfcRoad	HasObjectType	Se PredefinedType = .USERDEFINED., ObjectType tem de ser preenchido.

Datas como texto

Desde o IFC4 datas são strings ISO 8601: IfcDate ('2026-09-28'), IfcTime, IfcDateTime ('2026-09-28T10:00:00') e IfcDuration ('P2D'). As entidades de data do 2x3 (IfcCalendarDate, IfcLocalTime, IfcDateAndTime) não existem mais. IfcTimeStamp (inteiro Unix) continua no IfcOwnerHistory.

12. Arquivo completo comentado

Arquivo mínimo gerado com IfcOpenShell no esquema IFC4X3_ADD2 e validado sem erros (incluindo as regras WHERE): projeto, terreno, edifício, pavimento e uma parede com geometria, material em camadas, tipo e Pset.

```
ISO-10303-21;
HEADER;
FILE_DESCRIPTION(('ViewDefinition [ReferenceView]'),'2;1');
FILE_NAME('','2026-09-28T11:12:19',(''),(''),'IfcOpenShell 0.8.5-1c5b825','IfcOpenShell 0.8.5-1c5b825','');
FILE_SCHEMA(('IFC4X3_ADD2'));
ENDSEC;

DATA;
```

```
#1=IFCCARTESIANPOINT((0.,0.,0.));
#2=IFCAXIS2PLACEMENT3D(#1,$,$);
#3=IFCGEOMETRICREPRESENTATIONCONTEXT('Model','Model',3,1.0000000000000001E-05,#2,$);
#4=IFCGEOMETRICREPRESENTATIONSUBCONTEXT('Body','Model',*,*,*,*,#3,$,.MODEL_VIEW.,$);
#5=IFCSIUNIT(*,.LENGTHUNIT.,.MILLI.,.METRE.);
#6=IFCUNITASSIGNMENT((#5));
#7=IFCPROJECT('1EmGQM6KfAgP8pin0c8uLt',$,'Projeto Exemplo',$,$,$,$,(#3),#6);
#8=IFCAXIS2PLACEMENT3D(#1,$,$);
#9=IFCLOCALPLACEMENT($,#8);
#10=IFCSITE('22xCOgnDL9B8jiLBtP2vjR',$,'Terreno',$,$,#9,$,$,$,$,$,$,$,$);
#11=IFCAXIS2PLACEMENT3D(#1,$,$);
#12=IFCLOCALPLACEMENT(#9,#11);
#13=IFCBUILDING('1uK89Ygzz7HfbDyJ2fuFAV',$,'Edificio A',$,$,#12,$,$,$,$,$,$);
#14=IFCAXIS2PLACEMENT3D(#1,$,$);
#15=IFCLOCALPLACEMENT(#12,#14);
#16=IFCBUILDINGSTOREY('10S2sws9HA2x$NATI046Nj',$,'Terreo',$,$,#15,$,$,$,0.);
#17=IFCRELAGGREGATES('3r3ZTfP11CS8Ma2mrrnBwj',$,$,$,#7,(#10));
#18=IFCRELAGGREGATES('0_LbvIfub8VRzr1wwVzGvi',$,$,$,#10,(#13));
#19=IFCRELAGGREGATES('1PF9K7XyP2_h2SE2CZun_L',$,$,$,#13,(#16));
#20=IFCRECTANGLEPROFILEDEF(.AREA.,$,$,5000.,200.);
#21=IFCAXIS2PLACEMENT3D(#1,$,$);
#22=IFCDIRECTION((0.,0.,1.));
#23=IFCEXTRUDEDAREASOLID(#20,#21,#22,3000.);
#24=IFCSHAPEREPRESENTATION(#4,'Body','SweptSolid',(#23));
#25=IFCPRODUCTDEFINITIONSHAPE($,$,(#24));
#26=IFCAXIS2PLACEMENT3D(#1,$,$);
#27=IFCLOCALPLACEMENT(#15,#26);
#28=IFCWALL('1VISKZXADCF9jqhay1nExU',$,'PAR-01',$,$,#27,#25,$,.SOLIDWALL.);
#29=IFCWALLTYPE('1sTzRmsvTEEfmzDsY7aiLh',$,'Alvenaria 20cm',$,$,$,$,$,.SOLIDWALL.);
#30=IFCRELDEFINESBYTYPE('06KS89BGX3O8kIAmCMIIU6',$,$,$,(#28),#29);
#31=IFCMATERIAL('Bloco ceramico',$,$);
#32=IFCMATERIALLAYER(#31,200.,$,$,$,$,$);
#33=IFCMATERIALLAYERSET((#32),'Alvenaria 20cm',$);
#34=IFCMATERIALLAYERSETUSAGE(#33,.AXIS2.,.POSITIVE.,-100.,$);
#35=IFCRELASSOCIATESMATERIAL('0bCPUbjiv7yf40gYq$NDl6',$,$,$,(#28),#34);
#36=IFCRELCONTAINEDINSPATIALSTRUCTURE('0DybGSB$v5_x8kjT6_hVdj',$,$,$,(#28),#16);
#37=IFCPROPERTYSINGLEVALUE('IsExternal',$,IFCBOOLEAN(.T.),$);
#38=IFCPROPERTYSINGLEVALUE('FireRating',$,IFCLABEL('EI 60'),$);
#39=IFCPROPERTYSET('3hUaVL2D96HhxDhvKqiXrI',$,'Pset_WallCommon',$,(#37,#38));
#40=IFCRELDEFINESBYPROPERTIES('2PH1C4bqH3HRhl51FFzz7E',$,$,$,(#28),#39);
ENDSEC;
```

Como navegar este arquivo

- **Não há IfcOwnerHistory**: o 2º atributo de todas as entidades é $.
- **#3 e #4**: contexto geométrico 3D e subcontexto 'Body'. **#5–#6**: unidades (milímetro).
- **IFCSITE / IFCBUILDING / IFCBUILDINGSTOREY**: CompositionType é opcional ($). A árvore continua sendo montada por IFCRELAGGREGATES.
- **IFCRECTANGLEPROFILEDEF**: o Position do perfil é opcional.
- **IFCWALL**: 9 atributos; o último é PredefinedType (.SOLIDWALL.). O tipo IFCWALLTYPE é ligado por IFCRELDEFINESBYTYPE, acessível pelo inverso IsTypedBy.
- **IFCMATERIALLAYERSETUSAGE**: continua sendo a forma de dizer que a parede tem espessura constante definida por camadas (antes isso era a IfcWallStandardCase).
- **IFCPROPERTYSINGLEVALUE**: valores tipados porque NominalValue é um SELECT.

13. Novidades em relação ao 2x3 e entidades obsoletas

O 4.3 tem **343 entidades que não existiam no 2x3** (as introduzidas no IFC4 somadas às do 4.3). Distribuição por módulo:

Módulo	Novas	Exemplos
IfcProductExtension	42	IfcAlignment, IfcAlignmentCant, IfcAlignmentHorizontal, IfcAlignmentSegment, IfcAlignmentVertical, IfcBridge …
IfcHvacDomain	36	IfcAirTerminal, IfcAirTerminalBox, IfcAirToAirHeatRecovery, IfcBoiler, IfcBurner, IfcBurnerType …
IfcElectricalDomain	34	IfcAudioVisualAppliance, IfcAudioVisualApplianceType, IfcCableCarrierFitting, IfcCableCarrierSegment, IfcCableFitting, IfcCableFittingType …
IfcGeometryResource	34	IfcAxis2PlacementLinear, IfcBSplineCurveWithKnots, IfcBSplineSurface, IfcBSplineSurfaceWithKnots, IfcBoundaryCurve, IfcClothoid …
IfcGeometricModelResource	21	IfcAdvancedBrep, IfcAdvancedBrepWithVoids, IfcCartesianPointList, IfcCartesianPointList2D, IfcCartesianPointList3D, IfcDirectrixCurveSweptAreaSolid …
IfcSharedInfrastructureElements	18	IfcBorehole, IfcCourse, IfcCourseType, IfcEarthworksCut, IfcEarthworksElement, IfcEarthworksFill …
IfcStructuralElementsDomain	15	IfcCaissonFoundation, IfcCaissonFoundationType, IfcDeepFoundation, IfcDeepFoundationType, IfcFootingType, IfcPileType …
IfcKernel	15	IfcComplexPropertyTemplate, IfcContext, IfcPreDefinedPropertySet, IfcProjectLibrary, IfcPropertySetTemplate, IfcPropertyTemplate …
IfcSharedBldgElements	12	IfcBearing, IfcBearingType, IfcBuildingSystem, IfcChimney, IfcChimneyType, IfcDoorType …
IfcMaterialResource	11	IfcMaterialConstituent, IfcMaterialConstituentSet, IfcMaterialDefinition, IfcMaterialLayerWithOffsets, IfcMaterialProfile, IfcMaterialProfileSet …
IfcPortsAndWaterwaysDomain	10	IfcConveyorSegment, IfcConveyorSegmentType, IfcLiquidTerminal, IfcLiquidTerminalType, IfcMarineFacility, IfcMarinePart …
IfcDateTimeResource	9	IfcEventTime, IfcLagTime, IfcRecurrencePattern, IfcResourceTime, IfcSchedulingTime, IfcTaskTime …
IfcPresentationAppearanceResource	8	IfcColourRgbList, IfcIndexedColourMap, IfcIndexedPolygonalTextureMap, IfcIndexedTextureMap, IfcIndexedTriangleTextureMap, IfcTextureCoordinateIndices …
IfcRepresentationResource	8	IfcCoordinateOperation, IfcCoordinateReferenceSystem, IfcGeographicCRS, IfcMapConversion, IfcMapConversionScaled, IfcProjectedCRS …
IfcBuildingControlsDomain	7	IfcActuator, IfcAlarm, IfcController, IfcFlowInstrument, IfcSensor, IfcUnitaryControlElement …
IfcConstructionMgmtDomain	7	IfcConstructionEquipmentResourceType, IfcConstructionMaterialResourceType, IfcConstructionProductResourceType, IfcConstructionResourceType, IfcCrewResourceType, IfcLaborResourceType …
IfcGeometricConstraintResource	6	IfcAlignmentCantSegment, IfcAlignmentHorizontalSegment, IfcAlignmentParameterSegment, IfcAlignmentVerticalSegment, IfcConnectionVolumeGeometry, IfcLinearPlacement
IfcPlumbingFireProtectionDomain	6	IfcFireSuppressionTerminal, IfcInterceptor, IfcInterceptorType, IfcSanitaryTerminal, IfcStackTerminal, IfcWasteTerminal
IfcRailDomain	6	IfcRail, IfcRailType, IfcRailway, IfcRailwayPart, IfcTrackElement, IfcTrackElementType
IfcProcessExtension	5	IfcEvent, IfcEventType, IfcProcedureType, IfcTaskType, IfcWorkCalendar
IfcStructuralAnalysisDomain	5	IfcStructuralCurveAction, IfcStructuralCurveReaction, IfcStructuralLoadCase, IfcStructuralSurfaceAction, IfcStructuralSurfaceReaction
IfcRoadDomain	4	IfcKerb, IfcKerbType, IfcRoad, IfcRoadPart
IfcPropertyResource	3	IfcExtendedProperties, IfcPreDefinedProperties, IfcPropertyAbstraction
IfcExternalReferenceResource	3	IfcExternalInformation, IfcExternalReferenceRelationship, IfcResourceLevelRelationship
IfcStructuralLoadResource	3	IfcStructuralLoadConfiguration, IfcStructuralLoadOrResult, IfcSurfaceReinforcementArea
IfcSharedBldgServiceElements	2	IfcDistributionCircuit, IfcDistributionSystem
IfcSharedFacilitiesElements	2	IfcFurniture, IfcSystemFurnitureElement
IfcProfileResource	2	IfcMirroredProfileDef, IfcOpenCrossProfileDef
IfcConstraintResource	2	IfcReference, IfcResourceConstraintRelationship
IfcTopologyResource	1	IfcAdvancedFace

Módulo	Novas	Exemplos
IfcSharedComponentElements	1	IfcBuildingElementPartType
IfcMeasureResource	1	IfcConversionBasedUnitWithOffset
IfcPresentationDefinitionResource	1	IfcPresentationItem
IfcQuantityResource	1	IfcQuantityNumber
IfcApprovalResource	1	IfcResourceApprovalRelationship
IfcUtilityResource	1	IfcTableColumn

13.1 Entidades obsoletas (deprecated) que ainda existem no esquema

Estas entidades continuam válidas no IFC4X3_ADD2, então um arquivo que as usa passa na validação de esquema. Mas a documentação oficial pede que não sejam usadas em arquivos novos. Um conversor que apenas "casa nomes" vai mantê-las.

Entidade obsoleta	Desde	Usar no lugar
IfcWallStandardCase	IFC4.3	IfcWall + IfcMaterialLayerSetUsage
IfcBuildingSystem	IFC4.3.0.0	IfcBuiltSystem
IfcElectricDistributionBoard(Type)	IFC4.3.0.0	IfcDistributionBoard(Type)
IfcCivilElement(Type)	IFC4.3.0.0	Subtipos específicos de IfcBuiltElement, IfcEarthworksElement, IfcGeotechnicalElement
IfcRelConnectsPortToElement	IFC4.3.0.0	IfcRelNests (portas aninhadas no elemento)
IfcRelServicesBuildings	IFC4.3.0.0	IfcRelReferencedInSpatialStructure
IfcPostalAddress, IfcTelecomAddress	IFC4.3.0.0	Pset_Address
IfcTrapeziumProfileDef	IFC4.3.0.0	IfcArbitraryClosedProfileDef
IfcDoor/WindowLiningProperties, IfcDoor/WindowPanelProperties, IfcPermeableCoveringProperties	IFC4.3.2.0	Psets equivalentes (ex.: Pset_DoorLiningProperties)
IfcMaterialList	IFC4	IfcMaterialConstituentSet
IfcFlowSegment, IfcFlowFitting, IfcFlowTerminal, IfcFlowController, IfcEnergyConversionDevice... (instanciação)	IFC4	Subtipos específicos (IfcPipeSegment, IfcDuctFitting...)

> **Correção ao resumo anterior:** o resumo "IFC 4.3 — Visão e Estrutura" (seção 7.4) diz que IfcBuildingSystem e IfcElectricDistributionBoard foram renomeados. Na verdade os nomes novos (IfcBuiltSystem, IfcDistributionBoard) foram criados e os antigos **continuam no esquema como obsoletos**; verificado no IFC4X3_ADD2 do IfcOpenShell 0.8.5.

13.2 Renomeações e mudanças de estrutura relevantes

- IfcBuildingElement → **IfcBuiltElement** (e IfcBuildingElementType → IfcBuiltElementType).
- IfcBuilding passou a ser subtipo de **IfcFacility**.
- IfcProject passou a ser subtipo de **IfcContext** (ao lado de IfcProjectLibrary), com RepresentationContexts e UnitsInContext opcionais.
- IfcPile passou a ser subtipo de **IfcDeepFoundation** (ao lado de IfcCaissonFoundation).
- IfcGrid passou a ser subtipo de **IfcPositioningElement**.
- IfcRelVoidsElement e IfcRelProjectsElement passaram a ser IfcRelDecomposes; IfcRelAdheresToElement é nova.
- IfcElementQuantity passou a ser subtipo de IfcQuantitySet.
- PredefinedType está em 153 classes de ocorrência (no 2x3 eram 9).

14. Consultando o esquema por código

Tudo o que está neste documento pode ser consultado diretamente com o IfcOpenShell. Trecho usado para gerar as tabelas:

```
import ifcopenshell
import ifcopenshell.ifcopenshell_wrapper as W

s = W.schema_by_name("IFC4X3_ADD2")
e = s.declaration_by_name("IfcWall")

print(e.is_abstract(), e.supertype().name())          # abstrata? pai?
print([x.name() for x in e.subtypes()])              # filhos diretos
for i, a in enumerate(e.all_attributes()):           # ordem do STEP
    print(i, a.name(), a.type_of_attribute(), a.optional())
for inv in e.all_inverse_attributes():               # inversos
    print(inv.name(), inv.entity_reference().name(),
          inv.attribute_reference().name())

enum = s.declaration_by_name("IfcWallTypeEnum")
print(enum.enumeration_items())

# a partir de um arquivo aberto
m = ifcopenshell.open("modelo.ifc")
print(m.schema, m.schema_identifier)
decl = m.by_type("IfcWall")[0].is_a()                 # nome da classe
info = m.by_type("IfcWall")[0].wrapped_data.declaration().as_entity()
```

Apêndice: todas as entidades por módulo

Lista completa das entidades do esquema, agrupadas por camada e módulo. Entidades abstratas estão marcadas com (A). Tipos, enumerações e SELECTs foram omitidos.

Camada Domínio

IfcArchitectureDomain (5)

IfcDoorLiningProperties, IfcDoorPanelProperties, IfcPermeableCoveringProperties, IfcWindowLiningProperties, IfcWindowPanelProperties

IfcBuildingControlsDomain (12)

IfcActuator, IfcActuatorType, IfcAlarm, IfcAlarmType, IfcController, IfcControllerType, IfcFlowInstrument, IfcFlowInstrumentType, IfcSensor, IfcSensorType, IfcUnitaryControlElement, IfcUnitaryControlElementType

IfcConstructionMgmtDomain (14)

IfcConstructionEquipmentResource, IfcConstructionEquipmentResourceType, IfcConstructionMaterialResource, IfcConstructionMaterialResourceType, IfcConstructionProductResource, IfcConstructionProductResourceType, IfcConstructionResource (A), IfcConstructionResourceType (A), IfcCrewResource, IfcCrewResourceType, IfcLaborResource, IfcLaborResourceType, IfcSubContractResource, IfcSubContractResourceType

IfcElectricalDomain (50)

IfcAudioVisualAppliance, IfcAudioVisualApplianceType, IfcCableCarrierFitting, IfcCableCarrierFittingType, IfcCableCarrierSegment, IfcCableCarrierSegmentType, IfcCableFitting, IfcCableFittingType, IfcCableSegment, IfcCableSegmentType, IfcCommunicationsAppliance, IfcCommunicationsApplianceType, IfcDistributionBoard, IfcDistributionBoardType, IfcElectricAppliance, IfcElectricApplianceType, IfcElectricDistributionBoard, IfcElectricDistributionBoardType, IfcElectricFlowStorageDevice, IfcElectricFlowStorageDeviceType, IfcElectricFlowTreatmentDevice, IfcElectricFlowTreatmentDeviceType, IfcElectricGenerator, IfcElectricGeneratorType, IfcElectricMotor, IfcElectricMotorType, IfcElectricTimeControl, IfcElectricTimeControlType, IfcJunctionBox, IfcJunctionBoxType, IfcLamp, IfcLampType, IfcLightFixture, IfcLightFixtureType, IfcMobileTelecommunicationsAppliance, IfcMobileTelecommunicationsApplianceType, IfcMotorConnection, IfcMotorConnectionType, IfcOutlet, IfcOutletType, IfcProtectiveDevice, IfcProtectiveDeviceTrippingUnit, IfcProtectiveDeviceTrippingUnitType, IfcProtectiveDeviceType, IfcSolarDevice, IfcSolarDeviceType, IfcSwitchingDevice, IfcSwitchingDeviceType, IfcTransformer, IfcTransformerType

IfcHvacDomain (66)

IfcAirTerminal, IfcAirTerminalBox, IfcAirTerminalBoxType, IfcAirTerminalType, IfcAirToAirHeatRecovery, IfcAirToAirHeatRecoveryType, IfcBoiler, IfcBoilerType, IfcBurner, IfcBurnerType, IfcChiller, IfcChillerType, IfcCoil, IfcCoilType, IfcCompressor, IfcCompressorType, IfcCondenser, IfcCondenserType, IfcCooledBeam, IfcCooledBeamType, IfcCoolingTower, IfcCoolingTowerType, IfcDamper, IfcDamperType, IfcDuctFitting, IfcDuctFittingType, IfcDuctSegment, IfcDuctSegmentType, IfcDuctSilencer, IfcDuctSilencerType, IfcEngine, IfcEngineType, IfcEvaporativeCooler, IfcEvaporativeCoolerType, IfcEvaporator, IfcEvaporatorType, IfcFan, IfcFanType, IfcFilter, IfcFilterType, IfcFlowMeter, IfcFlowMeterType, IfcHeatExchanger, IfcHeatExchangerType, IfcHumidifier, IfcHumidifierType, IfcMedicalDevice, IfcMedicalDeviceType, IfcPipeFitting, IfcPipeFittingType, IfcPipeSegment, IfcPipeSegmentType, IfcPump, IfcPumpType, IfcSpaceHeater, IfcSpaceHeaterType, IfcTank, IfcTankType, IfcTubeBundle, IfcTubeBundleType, IfcUnitaryEquipment, IfcUnitaryEquipmentType, IfcValve, IfcValveType, IfcVibrationIsolator, IfcVibrationIsolatorType

IfcPlumbingFireProtectionDomain (10)

IfcFireSuppressionTerminal, IfcFireSuppressionTerminalType, IfcInterceptor, IfcInterceptorType, IfcSanitaryTerminal, IfcSanitaryTerminalType, IfcStackTerminal, IfcStackTerminalType, IfcWasteTerminal, IfcWasteTerminalType

IfcPortsAndWaterwaysDomain (10)

IfcConveyorSegment, IfcConveyorSegmentType, IfcLiquidTerminal, IfcLiquidTerminalType, IfcMarineFacility, IfcMarinePart, IfcMooringDevice, IfcMooringDeviceType, IfcNavigationElement, IfcNavigationElementType

IfcRailDomain (6)

IfcRail, IfcRailType, IfcRailway, IfcRailwayPart, IfcTrackElement, IfcTrackElementType

IfcRoadDomain (4)

IfcKerb, IfcKerbType, IfcRoad, IfcRoadPart

IfcStructuralAnalysisDomain (28)

IfcRelConnectsStructuralActivity, IfcRelConnectsStructuralMember, IfcRelConnectsWithEccentricity, IfcStructuralAction (A), IfcStructuralActivity (A), IfcStructuralAnalysisModel, IfcStructuralConnection (A), IfcStructuralCurveAction, IfcStructuralCurveConnection, IfcStructuralCurveMember, IfcStructuralCurveMemberVarying, IfcStructuralCurveReaction, IfcStructuralItem (A), IfcStructuralLinearAction, IfcStructuralLoadCase, IfcStructuralLoadGroup, IfcStructuralMember (A), IfcStructuralPlanarAction, IfcStructuralPointAction, IfcStructuralPointConnection, IfcStructuralPointReaction, IfcStructuralReaction (A), IfcStructuralResultGroup, IfcStructuralSurfaceAction, IfcStructuralSurfaceConnection, IfcStructuralSurfaceMember, IfcStructuralSurfaceMemberVarying, IfcStructuralSurfaceReaction

IfcStructuralElementsDomain (23)

IfcCaissonFoundation, IfcCaissonFoundationType, IfcDeepFoundation, IfcDeepFoundationType, IfcFooting, IfcFootingType, IfcPile, IfcPileType, IfcReinforcementDefinitionProperties, IfcReinforcingBar, IfcReinforcingBarType, IfcReinforcingElement (A), IfcReinforcingElementType (A), IfcReinforcingMesh, IfcReinforcingMeshType, IfcSurfaceFeature, IfcTendon, IfcTendonAnchor, IfcTendonAnchorType, IfcTendonConduit, IfcTendonConduitType, IfcTendonType, IfcVoidingFeature

Camada Interoperabilidade (Shared)

IfcSharedBldgElements (45)

IfcBeam, IfcBeamType, IfcBearing, IfcBearingType, IfcBuildingElementProxy, IfcBuildingElementProxyType, IfcBuildingSystem, IfcChimney, IfcChimneyType, IfcColumn, IfcColumnType, IfcCovering, IfcCoveringType, IfcCurtainWall, IfcCurtainWallType, IfcDoor, IfcDoorType, IfcMember, IfcMemberType, IfcPlate, IfcPlateType, IfcRailing, IfcRailingType, IfcRamp, IfcRampFlight, IfcRampFlightType, IfcRampType, IfcRelConnectsPathElements, IfcRelCoversBldgElements, IfcRelCoversSpaces, IfcRoof, IfcRoofType, IfcShadingDevice, IfcShadingDeviceType, IfcSlab, IfcSlabType, IfcStair, IfcStairFlight, IfcStairFlightType, IfcStairType, IfcWall, IfcWallStandardCase, IfcWallType, IfcWindow, IfcWindowType

IfcSharedBldgServiceElements (26)

IfcDistributionChamberElement, IfcDistributionChamberElementType, IfcDistributionCircuit, IfcDistributionControlElement, IfcDistributionControlElementType (A), IfcDistributionFlowElement, IfcDistributionFlowElementType (A), IfcDistributionPort, IfcDistributionSystem, IfcEnergyConversionDevice, IfcEnergyConversionDeviceType (A), IfcFlowController, IfcFlowControllerType (A), IfcFlowFitting, IfcFlowFittingType (A), IfcFlowMovingDevice, IfcFlowMovingDeviceType (A), IfcFlowSegment, IfcFlowSegmentType (A), IfcFlowStorageDevice, IfcFlowStorageDeviceType (A), IfcFlowTerminal, IfcFlowTerminalType (A), IfcFlowTreatmentDevice, IfcFlowTreatmentDeviceType (A), IfcRelFlowControlElements

IfcSharedComponentElements (10)

IfcBuildingElementPart, IfcBuildingElementPartType, IfcDiscreteAccessory, IfcDiscreteAccessoryType, IfcElementComponent (A), IfcElementComponentType (A), IfcFastener, IfcFastenerType, IfcMechanicalFastener, IfcMechanicalFastenerType

IfcSharedFacilitiesElements (7)

IfcAsset, IfcFurniture, IfcFurnitureType, IfcInventory, IfcOccupant, IfcSystemFurnitureElement, IfcSystemFurnitureElementType

IfcSharedInfrastructureElements (18)

IfcBorehole, IfcCourse, IfcCourseType, IfcEarthworksCut, IfcEarthworksElement, IfcEarthworksFill, IfcGeomodel, IfcGeoslice, IfcGeotechnicalAssembly (A), IfcGeotechnicalElement (A), IfcGeotechnicalStratum, IfcPavement, IfcPavementType, IfcReinforcedSoil, IfcSign, IfcSignType, IfcSignal, IfcSignalType

IfcSharedMgmtElements (5)

IfcActionRequest, IfcCostItem, IfcCostSchedule, IfcPermit, IfcProjectOrder

Camada Núcleo (Core)

IfcControlExtension (3)

IfcPerformanceHistory, IfcRelAssociatesApproval, IfcRelAssociatesConstraint

IfcKernel (49)

IfcActor, IfcComplexPropertyTemplate, IfcContext (A), IfcControl (A), IfcGroup, IfcObject (A), IfcObjectDefinition (A), IfcPreDefinedPropertySet (A), IfcProcess (A), IfcProduct (A), IfcProject, IfcProjectLibrary, IfcPropertyDefinition (A), IfcPropertySet, IfcPropertySetDefinition (A), IfcPropertySetTemplate, IfcPropertyTemplate (A), IfcPropertyTemplateDefinition (A), IfcQuantitySet (A), IfcRelAggregates, IfcRelAssigns (A), IfcRelAssignsToActor, IfcRelAssignsToControl, IfcRelAssignsToGroup, IfcRelAssignsToGroupByFactor, IfcRelAssignsToProcess, IfcRelAssignsToProduct, IfcRelAssignsToResource, IfcRelAssociates (A), IfcRelAssociatesClassification, IfcRelAssociatesDocument, IfcRelAssociatesLibrary, IfcRelConnects (A), IfcRelDeclares, IfcRelDecomposes (A), IfcRelDefines (A), IfcRelDefinesByObject, IfcRelDefinesByProperties, IfcRelDefinesByTemplate, IfcRelDefinesByType, IfcRelNests, IfcRelationship (A), IfcResource (A), IfcRoot (A), IfcSimplePropertyTemplate, IfcTypeObject, IfcTypeProcess (A), IfcTypeProduct, IfcTypeResource (A)

IfcProcessExtension (11)

IfcEvent, IfcEventType, IfcProcedure, IfcProcedureType, IfcRelSequence, IfcTask, IfcTaskType, IfcWorkCalendar, IfcWorkControl (A), IfcWorkPlan, IfcWorkSchedule

IfcProductExtension (82)

IfcAlignment, IfcAlignmentCant, IfcAlignmentHorizontal, IfcAlignmentSegment, IfcAlignmentVertical, IfcAnnotation, IfcBridge, IfcBridgePart, IfcBuilding, IfcBuildingStorey, IfcBuiltElement, IfcBuiltElementType, IfcBuiltSystem, IfcCivilElement, IfcCivilElementType, IfcDistributionElement, IfcDistributionElementType, IfcElement (A), IfcElementAssembly, IfcElementAssemblyType, IfcElementQuantity, IfcElementType (A), IfcExternalSpatialElement, IfcExternalSpatialStructureElement (A), IfcFacility, IfcFacilityPart (A), IfcFacilityPartCommon, IfcFeatureElement (A), IfcFeatureElementAddition (A), IfcFeatureElementSubtraction (A), IfcFurnishingElement, IfcFurnishingElementType, IfcGeographicElement, IfcGeographicElementType, IfcGrid, IfcImpactProtectionDevice, IfcImpactProtectionDeviceType, IfcLinearElement, IfcLinearPositioningElement, IfcOpeningElement, IfcPort (A), IfcPositioningElement (A), IfcProjectionElement, IfcReferent, IfcRelAdheresToElement, IfcRelAssociatesMaterial, IfcRelAssociatesProfileDef, IfcRelConnectsElements, IfcRelConnectsPortToElement, IfcRelConnectsPorts, IfcRelConnectsWithRealizingElements, IfcRelContainedInSpatialStructure, IfcRelFillsElement, IfcRelInterferesElements, IfcRelPositions, IfcRelProjectsElement, IfcRelReferencedInSpatialStructure, IfcRelServicesBuildings, IfcRelSpaceBoundary, IfcRelSpaceBoundary1stLevel, IfcRelSpaceBoundary2ndLevel, IfcRelVoidsElement, IfcSite, IfcSpace, IfcSpaceType, IfcSpatialElement (A), IfcSpatialElementType (A), IfcSpatialStructureElement (A), IfcSpatialStructureElementType (A), IfcSpatialZone, IfcSpatialZoneType, IfcSystem, IfcTransportElement, IfcTransportElementType, IfcTransportationDevice (A), IfcTransportationDeviceType (A), IfcVehicle, IfcVehicleType, IfcVibrationDamper, IfcVibrationDamperType, IfcVirtualElement, IfcZone

Camada Recursos

IfcActorResource (8)

IfcActorRole, IfcAddress (A), IfcOrganization, IfcOrganizationRelationship, IfcPerson, IfcPersonAndOrganization, IfcPostalAddress, IfcTelecomAddress

IfcApprovalResource (3)

IfcApproval, IfcApprovalRelationship, IfcResourceApprovalRelationship

IfcConstraintResource (5)

IfcConstraint (A), IfcMetric, IfcObjective, IfcReference, IfcResourceConstraintRelationship

IfcCostResource (3)

IfcAppliedValue, IfcCostValue, IfcCurrencyRelationship

IfcDateTimeResource (14)

IfcEventTime, IfcIrregularTimeSeries, IfcIrregularTimeSeriesValue, IfcLagTime, IfcRecurrencePattern, IfcRegularTimeSeries, IfcResourceTime, IfcSchedulingTime (A), IfcTaskTime, IfcTaskTimeRecurring, IfcTimePeriod, IfcTimeSeries (A), IfcTimeSeriesValue, IfcWorkTime

IfcExternalReferenceResource (11)

IfcClassification, IfcClassificationReference, IfcDocumentInformation, IfcDocumentInformationRelationship, IfcDocumentReference, IfcExternalInformation (A), IfcExternalReference (A), IfcExternalReferenceRelationship, IfcLibraryInformation, IfcLibraryReference, IfcResourceLevelRelationship (A)

IfcGeometricConstraintResource (16)

IfcAlignmentCantSegment, IfcAlignmentHorizontalSegment, IfcAlignmentParameterSegment (A), IfcAlignmentVerticalSegment, IfcConnectionCurveGeometry, IfcConnectionGeometry (A), IfcConnectionPointEccentricity, IfcConnectionPointGeometry, IfcConnectionSurfaceGeometry, IfcConnectionVolumeGeometry, IfcGridAxis, IfcGridPlacement, IfcLinearPlacement, IfcLocalPlacement, IfcObjectPlacement (A), IfcVirtualGridIntersection

IfcGeometricModelResource (48)

IfcAdvancedBrep, IfcAdvancedBrepWithVoids, IfcBlock, IfcBooleanClippingResult, IfcBooleanResult, IfcBoundingBox, IfcBoxedHalfSpace, IfcCartesianPointList (A), IfcCartesianPointList2D, IfcCartesianPointList3D, IfcCsgPrimitive3D (A), IfcCsgSolid, IfcDirectrixCurveSweptAreaSolid (A), IfcDirectrixDerivedReferenceSweptAreaSolid, IfcExtrudedAreaSolid, IfcExtrudedAreaSolidTapered, IfcFaceBasedSurfaceModel, IfcFacetedBrep, IfcFacetedBrepWithVoids, IfcFixedReferenceSweptAreaSolid, IfcGeometricCurveSet, IfcGeometricSet, IfcHalfSpaceSolid, IfcIndexedPolygonalFace, IfcIndexedPolygonalFaceWithVoids, IfcManifoldSolidBrep (A), IfcPolygonalBoundedHalfSpace, IfcPolygonalFaceSet, IfcRectangularPyramid, IfcRevolvedAreaSolid, IfcRevolvedAreaSolidTapered, IfcRightCircularCone, IfcRightCircularCylinder, IfcSectionedSolid (A), IfcSectionedSolidHorizontal, IfcSectionedSpine, IfcSectionedSurface, IfcShellBasedSurfaceModel, IfcSolidModel (A), IfcSphere, IfcSurfaceCurveSweptAreaSolid, IfcSweptAreaSolid (A), IfcSweptDiskSolid, IfcSweptDiskSolidPolygonal, IfcTessellatedFaceSet (A), IfcTessellatedItem (A), IfcTriangulatedFaceSet, IfcTriangulatedIrregularNetwork

IfcGeometryResource (75)

IfcAxis1Placement, IfcAxis2Placement2D, IfcAxis2Placement3D, IfcAxis2PlacementLinear, IfcBSplineCurve (A), IfcBSplineCurveWithKnots, IfcBSplineSurface (A), IfcBSplineSurfaceWithKnots, IfcBoundaryCurve, IfcBoundedCurve (A), IfcBoundedSurface (A), IfcCartesianPoint, IfcCartesianTransformationOperator (A), IfcCartesianTransformationOperator2D, IfcCartesianTransformationOperator2DnonUniform, IfcCartesianTransformationOperator3D, IfcCartesianTransformationOperator3DnonUniform, IfcCircle, IfcClothoid, IfcCompositeCurve, IfcCompositeCurveOnSurface, IfcCompositeCurveSegment, IfcConic (A), IfcCosineSpiral, IfcCurve (A), IfcCurveBoundedPlane, IfcCurveBoundedSurface, IfcCurveSegment, IfcCylindricalSurface, IfcDirection, IfcElementarySurface (A), IfcEllipse, IfcGeometricRepresentationItem (A), IfcGradientCurve, IfcIndexedPolyCurve, IfcIntersectionCurve, IfcLine, IfcMappedItem, IfcOffsetCurve (A), IfcOffsetCurve2D, IfcOffsetCurve3D, IfcOffsetCurveByDistances, IfcOuterBoundaryCurve, IfcPcurve, IfcPlacement (A), IfcPlane, IfcPoint (A), IfcPointByDistanceExpression, IfcPointOnCurve, IfcPointOnSurface, IfcPolyline, IfcPolynomialCurve, IfcRationalBSplineCurveWithKnots, IfcRationalBSplineSurfaceWithKnots, IfcRectangularTrimmedSurface, IfcReparametrisedCompositeCurveSegment, IfcRepresentationItem (A), IfcRepresentationMap, IfcSeamCurve, IfcSecondOrderPolynomialSpiral, IfcSegment (A), IfcSegmentedReferenceCurve, IfcSeventhOrderPolynomialSpiral, IfcSineSpiral, IfcSphericalSurface, IfcSpiral (A), IfcSurface (A), IfcSurfaceCurve, IfcSurfaceOfLinearExtrusion, IfcSurfaceOfRevolution, IfcSweptSurface (A), IfcThirdOrderPolynomialSpiral, IfcToroidalSurface, IfcTrimmedCurve, IfcVector

IfcMaterialResource (18)

IfcMaterial, IfcMaterialClassificationRelationship, IfcMaterialConstituent, IfcMaterialConstituentSet, IfcMaterialDefinition (A), IfcMaterialLayer, IfcMaterialLayerSet, IfcMaterialLayerSetUsage, IfcMaterialLayerWithOffsets, IfcMaterialList, IfcMaterialProfile, IfcMaterialProfileSet, IfcMaterialProfileSetUsage, IfcMaterialProfileSetUsageTapering, IfcMaterialProfileWithOffsets, IfcMaterialProperties, IfcMaterialRelationship, IfcMaterialUsageDefinition (A)

IfcMeasureResource (11)

IfcContextDependentUnit, IfcConversionBasedUnit, IfcConversionBasedUnitWithOffset, IfcDerivedUnit, IfcDerivedUnitElement, IfcDimensionalExponents, IfcMeasureWithUnit, IfcMonetaryUnit, IfcNamedUnit (A), IfcSIUnit, IfcUnitAssignment

IfcPresentationAppearanceResource (46)

IfcBlobTexture, IfcColourRgb, IfcColourRgbList, IfcColourSpecification (A), IfcCurveStyle, IfcCurveStyleFont, IfcCurveStyleFontAndScaling, IfcCurveStyleFontPattern, IfcDraughtingPreDefinedColour, IfcDraughtingPreDefinedCurveFont, IfcExternallyDefinedHatchStyle, IfcExternallyDefinedSurfaceStyle, IfcExternallyDefinedTextFont, IfcFillAreaStyle, IfcFillAreaStyleHatching, IfcFillAreaStyleTiles, IfcImageTexture, IfcIndexedColourMap, IfcIndexedPolygonalTextureMap, IfcIndexedTextureMap (A), IfcIndexedTriangleTextureMap, IfcPixelTexture, IfcPreDefinedColour (A), IfcPreDefinedCurveFont (A), IfcPreDefinedItem (A), IfcPreDefinedTextFont (A), IfcPresentationStyle (A), IfcStyledItem, IfcSurfaceStyle, IfcSurfaceStyleLighting, IfcSurfaceStyleRefraction, IfcSurfaceStyleRendering, IfcSurfaceStyleShading, IfcSurfaceStyleWithTextures, IfcSurfaceTexture (A), IfcTextStyle, IfcTextStyleFontModel, IfcTextStyleForDefinedFont, IfcTextStyleTextModel, IfcTextureCoordinate (A), IfcTextureCoordinateGenerator, IfcTextureCoordinateIndices, IfcTextureCoordinateIndicesWithVoids, IfcTextureMap, IfcTextureVertex, IfcTextureVertexList

IfcPresentationDefinitionResource (6)

IfcAnnotationFillArea, IfcPlanarBox, IfcPlanarExtent, IfcPresentationItem (A), IfcTextLiteral, IfcTextLiteralWithExtent

IfcPresentationOrganizationResource (10)

IfcLightDistributionData, IfcLightIntensityDistribution, IfcLightSource (A), IfcLightSourceAmbient, IfcLightSourceDirectional, IfcLightSourceGoniometric, IfcLightSourcePositional, IfcLightSourceSpot, IfcPresentationLayerAssignment, IfcPresentationLayerWithStyle

IfcProfileResource (28)

IfcArbitraryClosedProfileDef, IfcArbitraryOpenProfileDef, IfcArbitraryProfileDefWithVoids, IfcAsymmetricIShapeProfileDef, IfcCShapeProfileDef, IfcCenterLineProfileDef, IfcCircleHollowProfileDef, IfcCircleProfileDef, IfcCompositeProfileDef, IfcDerivedProfileDef, IfcEllipseProfileDef, IfcIShapeProfileDef, IfcLShapeProfileDef, IfcMirroredProfileDef, IfcOpenCrossProfileDef, IfcParameterizedProfileDef (A), IfcProfileDef, IfcProfileProperties, IfcRectangleHollowProfileDef, IfcRectangleProfileDef, IfcReinforcementBarProperties, IfcRoundedRectangleProfileDef, IfcSectionProperties, IfcSectionReinforcementProperties, IfcTShapeProfileDef, IfcTrapeziumProfileDef, IfcUShapeProfileDef, IfcZShapeProfileDef

IfcPropertyResource (14)

IfcComplexProperty, IfcExtendedProperties (A), IfcPreDefinedProperties (A), IfcProperty (A), IfcPropertyAbstraction (A), IfcPropertyBoundedValue, IfcPropertyDependencyRelationship, IfcPropertyEnumeratedValue, IfcPropertyEnumeration, IfcPropertyListValue, IfcPropertyReferenceValue, IfcPropertySingleValue, IfcPropertyTableValue, IfcSimpleProperty (A)

IfcQuantityResource (10)

IfcPhysicalComplexQuantity, IfcPhysicalQuantity (A), IfcPhysicalSimpleQuantity (A), IfcQuantityArea, IfcQuantityCount, IfcQuantityLength, IfcQuantityNumber, IfcQuantityTime, IfcQuantityVolume, IfcQuantityWeight

IfcRepresentationResource (21)

IfcCoordinateOperation (A), IfcCoordinateReferenceSystem (A), IfcGeographicCRS, IfcGeometricRepresentationContext, IfcGeometricRepresentationSubContext, IfcMapConversion, IfcMapConversionScaled, IfcMaterialDefinitionRepresentation, IfcProductDefinitionShape, IfcProductRepresentation (A), IfcProjectedCRS, IfcRepresentation (A), IfcRepresentationContext (A), IfcRigidOperation, IfcShapeAspect, IfcShapeModel (A), IfcShapeRepresentation, IfcStyleModel (A), IfcStyledRepresentation, IfcTopologyRepresentation, IfcWellKnownText

IfcStructuralLoadResource (20)

IfcBoundaryCondition (A), IfcBoundaryEdgeCondition, IfcBoundaryFaceCondition, IfcBoundaryNodeCondition, IfcBoundaryNodeConditionWarping, IfcFailureConnectionCondition, IfcSlippageConnectionCondition, IfcStructuralConnectionCondition (A), IfcStructuralLoad (A), IfcStructuralLoadConfiguration, IfcStructuralLoadLinearForce, IfcStructuralLoadOrResult (A), IfcStructuralLoadPlanarForce, IfcStructuralLoadSingleDisplacement, IfcStructuralLoadSingleDisplacementDistortion, IfcStructuralLoadSingleForce, IfcStructuralLoadSingleForceWarping, IfcStructuralLoadStatic (A), IfcStructuralLoadTemperature, IfcSurfaceReinforcementArea

IfcTopologyResource (20)

IfcAdvancedFace, IfcClosedShell, IfcConnectedFaceSet, IfcEdge, IfcEdgeCurve, IfcEdgeLoop, IfcFace, IfcFaceBound, IfcFaceOuterBound, IfcFaceSurface, IfcLoop, IfcOpenShell, IfcOrientedEdge, IfcPath, IfcPolyLoop, IfcSubedge, IfcTopologicalRepresentationItem (A), IfcVertex, IfcVertexLoop, IfcVertexPoint

IfcUtilityResource (5)

IfcApplication, IfcOwnerHistory, IfcTable, IfcTableColumn, IfcTableRow